

Shirshendu Ranjana Tripathi

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EDUCATION

Indian Institute of Information Technology Design and Manufacturing

B.Tech in Electronics and Communication Engineering

Jabalpur, India

2024 – 2028

Metropolitan School

ICSE (Class X) | ISC (Class XII)

Gorakhpur, Uttar Pradesh

TECHNICAL SKILLS

Languages: C++, Python, Java, JavaScript, TypeScript

CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming (OOP), System Design, REST APIs

Web Development: React, Next.js 14, Node.js, Express.js, Django, Flask, RESTful APIs

Databases: MongoDB, PostgreSQL, SQL, Vector Databases

AI / ML: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), NLP, Computer Vision, OCR Pipelines, Prompt Engineering

Tools & Platforms: Git, GitHub, Docker, Figma, VS Code, Postman, Linux, Streamlit, OSINT APIs

PROJECTS

Saathi | *Next.js 14, Node.js, MongoDB, ElevenLabs TTS, Twilio, Gemini LLM* | [GitHub](#)

2026

- **Won ElevenLabs Sponsor Track** at MLH HackByte 4.0 (400+ participants); architected an AI platform that places proactive Hindi voice calls to elderly users via Twilio—no smartphone required—pairing ElevenLabs TTS with Gemini LLM to hold context-aware, emotionally adaptive conversations.
- Shipped a family monitoring dashboard (Next.js 14, Express.js, MongoDB) with real-time mood tracking, risk-flagging, and scheduled call management; **live in production within 48 hours** of the hackathon closing.
- Cut caregiver response time by **~60%** by replacing manual check-ins with automated safety alerts routed through a centralized interface.

TrueSignal | *Python, Groq LLaMA-3, Sentence-BERT, RAG, Streamlit, REST APIs* | [GitHub](#)

2026

- Developed a talent-verification platform that cross-references resume claims against live GitHub, LeetCode, and Codeforces data via OSINT APIs, **slashing credential-fraud screening time by ~80%**.
- Composed a multi-layer scoring pipeline—Cosine Similarity (Sentence-BERT) for semantic match, Shannon Entropy for language diversity, Jaccard Index for skill overlap—then layered Groq LLaMA-3 on top to generate plain-English audit reports and a RAG talent pool with sub-second vector retrieval.
- Attained **92%+ precision** in detecting inflated credentials across 200+ synthetic candidate profiles during validation.

CureCode | *Python, OCR, RAG, LLMs, Vector Search* | [GitHub](#)

2025

- Constructed an insurance-document analysis tool that extracts structured data from dense policy PDFs via OCR, then exposes it through a RAG interface—letting users query 100+ page documents in plain language.
- **National finalist, Hack The Future 2025** (IIT Gandhinagar) — **top 15 of 300+ teams**; system cut manual policy-review time by an estimated **75%**.

Streamify | *React, Node.js, Express.js, MongoDB, REST APIs* | [GitHub](#)

2024

- Launched a music streaming web app with JWT authentication, real-time search, and a RESTful backend wired to external music APIs for live metadata and playback.
- Structured the codebase for horizontal scale—**500+ concurrent sessions** sustained with sub-200ms API response times across a fully responsive React UI.

ACHIEVEMENTS & LEADERSHIP

MLH HackByte 4.0 | ElevenLabs Sponsor Track Winner — delivered Saathi with team SillyCoders in under 36 hours

2026

NIDAR Disaster Management Challenge | 7th nationally out of 350+ teams

2026

GirlScript Summer of Code | Top 50 Mentor nationally — mentored 30+ contributors across a 3-month open-source programme

2025

CodeChef | [3-Star rating](#) | [50+ public repositories](#) on GitHub

Aero Fabrication Club, IIITDM Jabalpur | Incoming Co-coordinator, 100+ member club

CERTIFICATIONS

CS50x — Introduction to Computer Science, Harvard University

AWS Educate — Introduction to Generative AI

Postman API Fundamentals Student Expert

RELEVANT COURSEWORK

Data Structures & Algorithms | Signal Processing | Analog Electronics | Digital Electronics | Computer Networks | Object-Oriented Programming